## Exam AWS Certified Developer - Associate DVA-C02 All Questions

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## EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 507
Topic #: 1
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A developer has implemented an AWS Lambda function that inserts new customers into an Amazon RDS database. The function is expected to run hundreds of times each hour. The function and RDS database are in the same VPC. The function is configured to use 512 MB of RAM and is based on the following pseudo code:

```
def lambda_handler (event, context):

      db = database.connect()
      db.statement('INSERT INTO Customers (CustomerName) VALUES (event.name)')
      db.execute()
      db.close()
```

After successfully testing the function multiple times, the developer notices that the execution time is longer than expected.

What should the developer do to improve performance?

A. Increase the reserved concurrency of the Lambda function.

B. Increase the size of the RDS database to facilitate an increased number of database connections each hour.

C. Move the database connection and close statement out of the handler. Place the connection in the global space.

D. Replace Amazon RDS with Amazon DynamoDB to implement control over the number of writes per second.

Show Suggested Answer

by e886835 at *Feb. 6, 2025, 9:17 a.m.*

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## Comments

**fsk8** 7 months, 3 weeks ago

**Selected Answer: C**

A - Concurrency helps scale, but doesn't reduce individual function execution time.
B - Database scaling is not likely the primary fix for individual slow Lambda. No info on database sizing or other lambdas running slow so eliminated this as a solution.
D - Changing the database type is an architectural shift, not a performance tune.

C feels correct - Reusing connections globally prevents connection overhead on subsequent invocations.

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**0bdf3af** 1 year, 1 month ago

**Selected Answer: C**

connection to database as global variable outside the handler

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**LingZ** 1 year, 1 month ago

**Selected Answer: C**

The best answer is C: Move the database connection and close statement out of the handler. Place the connection in the global space. This solution would:

Create the connection once when the Lambda container initializes
Reuse that same connection across multiple invocations
Significantly reduce the overhead of establishing new connections
Improve execution time and resource usage
Follow AWS best practices for Lambda-RDS connections

upvoted 2 times

**e886835** 1 year, 1 month ago

**Selected Answer: A**

Reserved concurrency defines the maximum number of concurrent executions for a Lambda function. If the function is expected to run

hundreds of times each hour, it could be hitting concurrency limits, causing delays due to Lambda being throttled, which would lead to longer execution times.

upvoted 1 times